UCF Physics: AST 5765/4762: (Advanced) Astronomical Data Analysis
Practicum 5: Combining Frames and Infrared Sky Correction
10/16/2025

Resources:
1. Chapter 6 of Howell
2. Appendix C of Howell

This assignment is the start of HW8. It continues the correction of the data from HW6.

1. (**10 points**) Copy the routines and homework files from HW7 into the directory for this assignment (`hw8_yourname`). Correct any errors you may have had, making a comment that says "# FIXED: " and giving the date.

2. (**10 points**) Write a function called `normmedcomb()` that implements the normalized median combination method.
The function should take as input:
- a 3D array of data that contains the dark-subtracted sky flux (e.g., from your HW6 dataset)
- an *optional keyword* that contains a tuple with the corner coordinates of the normalization region (e.g., `((y1, x1), (y2, x2))`). The default region for the normalization should be the whole image (the whole 2D array). Note that the normalization region is the region from which we calculate the normalization factor in each frame. The routine should then return a tuple containing:

    (a) the normalized, median-combined, 2D array, and
    (b) a 1D array containing the normalization factors.

    Be sure not to modify the 3D input array in the caller!

3. (**10 points**) Call routine `normmedcomb()` on your dark-subtracted object data from HW6. Use the following normalization region:

```
((225, 225), (-225, -225))   # ((y1, x1), (y2, x2))
```

This large normalization region avoids obviously-bad areas of the array (and if everyone uses the same one, grading is easier!).

Assign the result to a variable. You will use it throughout the rest of the homeworks (e.g., HW9).

Write the results to a FITS file named `sky_13s_mednorm.fits`.

Print the normalization factors and include them in a comment or dummy string in your main homework file.